CSCI 232 Assignment 03 – C Pointers & Node Fundamentals

Points: 10

Instructions

1. **Fork** the repository on GitHub: `https://github.com/jpach-cs/232-ass03-template`
2. **Clone** your forked repository to your local machine.
3. Open `src/code.c` and `src/tests.c`.
4. Uncomment the author block at the top of `src/code.c` and fill in your name:

```
// char *AUTHOR_NAME       = "Your Name";
// char *AUTHOR_AUTHORSHIP = "I acknowledge that I have worked on this
// assignment independently...";
```

5. Work through each function in **two parts**:

Part 1 — `src/code.c`

- Read the description and hint for each function.
- Implement the function body.
- Do **not** modify function signatures.

Part 2 — `src/tests.c`

- Read the description for each test function.
- Implement the test body using `TEST_ASSERT_`.
- Use only **stack variables** — no `malloc`.
- Do **not** modify function names or signatures.

6. When all local tests pass — **commit** and **push** to your repository.
7. Go to the **Actions** tab on GitHub and verify the CI results.

Functions to implement — `src/code.c`

Function	Description
`swap`	Swap values of two integers via pointers
`find_last_char`	Return pointer to last character before `\0`
`link_three`	Connect three nodes: `a -> b -> c -> NULL`
`remove_middle`	Reconnect `a -> c`, disconnect `b`
`remove_last`	Set `b->nextPtr = NULL`

Function	Description
`remove_first`	Set `a->nextPtr = NULL`
`swap_ptrs`	Swap two pointers using double pointers
`nullify`	Set pointer to NULL via double pointer
`assign_bytes`	Assign bytes 1..8 via cast pointer
`sum_chain`	Return sum of all values in chain

Tests to implement — `src/tests.c`

Test function	What to verify
`test_swap_basic`	Two different values are exchanged
`test_swap_equal`	Equal values remain unchanged
`test_find_last_basic`	Correct pointer to last character
`test_find_last_single`	Single character string
`test_find_last_empty`	Empty string returns NULL
`test_link_three_chain`	All three pointers correctly set
`test_remove_middle_reconnects`	`a->nextPtr == &c`, `b->nextPtr == NULL`
`test_remove_last_truncates`	`b->nextPtr == NULL`
`test_remove_first_isolates`	`a->nextPtr == NULL`, rest intact
`test_swap_ptrs_redirects`	Each pointer points to the other variable
`test_swap_ptrs_values_unchanged`	Original integer values not modified
`test_nullify_sets_null`	Pointer is NULL after call
`test_nullify_value_unchanged`	Original value not modified
`test_assign_bytes_first_and_last`	Byte 0 == 1, byte 7 == 8
`test_assign_bytes_all`	Every byte i contains i+1
`test_sum_chain_basic`	Correct sum for three nodes
`test_sum_chain_single`	Single node returns its value
`test_sum_chain_null`	NULL returns 0

Rules

- Work on **stack only** — no `malloc`, no `free`
- Do **not** modify function signatures in `code.c` or `tests.c`
- CI runs your `tests.c` together with instructor tests — both must pass

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Grading

Component	Points
All functions correctly implemented	5
All test functions correctly written	5
Total	**10**

Author block — required

```
char *AUTHOR_NAME       = "Your Name";
char *AUTHOR_AUTHORSHIP = "..."
```